LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	System protection of AFCS, explanation of comparator, rate trigger system, duplex system, cross coupled feedback system, triplex system, quadruplex and model following system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	System protection of AFCS, explanation of comparator, rate trigger system, duplex system, cross coupled feedback system, triplex system, quadruplex and model following system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and redundancy configurations; Safety: No live electrical connections; Instructor supervision: Monitor diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an AFCS?
(b) Define the term 'redundancy' in avionics.
(c) List three common failure modes in flight control systems.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Comparator function and signal comparison • Rate trigger system operation principles • Duplex system architecture and redundancy	L & Ds	PPT, IB	15 min
(b)	• Cross coupled feedback system mechanics • Triplex system voting logic and reliability • Quadruplex system architecture overview	L & Ds	PPT, PDF	20 min
(c)	• Model following system concept and application • System protection strategies summary • Comparison of redundancy levels	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a triplex system detects a faulty channel.
(b) Describe the role of the rate trigger in system protection.
(c) Differentiate between duplex and quadruplex redundancy.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of system protection discussed in this lesson apply to the position and speed control servomechanisms covered in LP 21.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Position control servomechanism, speed control servomechanism.	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min